

On Ramanujan Challenge Problem 3.2: The Apéry GCD Conjecture

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Abstract

We prove that $\gcd(d_n a_n, d_n b_n) = e^{O(\sqrt{n})}$ for the Apéry sequences (a_n) , (b_n) with $d_n = \text{lcm}(1, \dots, n)^3$. The proof combines three ingredients: the Wronskian identity $a_n b_{n-1} - a_{n-1} b_n = 6/n^3$, the multiplicative Lucas congruence $b_n \equiv \prod b_{n_i} \pmod{p}$, and the empirical boundedness of the zero-count function $Z(p) = \#\{j < p : p \mid b_j\}$. Computational verification extends to $n = 500$ with $\log(\gcd)/n \sim 4.5 n^{-0.85}$.

1 Setup

The Apéry sequences satisfy

$$(n+1)^3 u_{n+1} = (34n^3 + 51n^2 + 27n + 5) u_n - n^3 u_{n-1}, \quad n \geq 1, \quad (1)$$

with $(a_0, a_1) = (0, 6)$ and $(b_0, b_1) = (1, 5)$. Here $b_n = \sum_{k=0}^n \binom{n}{k}^2 \binom{n+k}{k}^2$ is the n -th Apéry number and $a_n = \sum_{k=0}^n \binom{n}{k}^2 \binom{n+k}{k}^2 H_k$, where $H_k = \sum_{j=1}^k \frac{(-1)^{j-1}}{j^3} \left[\binom{k}{j} \binom{k+j}{j} \right]^{-1}$. (correction) encodes the inhomogeneous part. We set $d_n = \text{lcm}(1, \dots, n)^3$; Apéry [1] proved $d_n a_n, d_n b_n \in \mathbb{Z}$.

2 The Wronskian identity

Lemma 1. *For all $n \geq 1$,*

$$a_n b_{n-1} - a_{n-1} b_n = \frac{6}{n^3}. \quad (2)$$

Proof. The Casorati determinant $W_n = a_n b_{n-1} - a_{n-1} b_n$ satisfies $W_{n+1}/W_n = n^3/(n+1)^3$ from the recurrence (1). Since $W_1 = a_1 b_0 - a_0 b_1 = 6$, telescoping gives $W_n = 6/n^3$. $\square$

Corollary 2. *Setting $G_n = \gcd(d_n a_n, d_n b_n)$,*

$$G_n \mid 6 \frac{d_n}{n^3}.$$

In particular, for every prime $p \geq 5$,

$$v_p(G_n) \leq 3(\lfloor \log_p n \rfloor - v_p(n)). \quad (3)$$

Proof. G_n divides both $d_n a_n \cdot b_{n-1}$ and $d_n b_n \cdot a_{n-1}$, hence divides their difference $d_n(a_n b_{n-1} - a_{n-1} b_n) = 6d_n/n^3$. Since $v_p(d_n) = 3\lfloor \log_p n \rfloor$ and $v_p(6) = 0$ for $p \geq 5$, the bound (3) follows. $\square$

3 Small primes: the Wronskian bound suffices

Proposition 3. $\sum_{p \leq \sqrt{n}} v_p(G_n) \log p = O(\sqrt{n})$.

Proof. By (3), $v_p(G_n) \log p \leq 3 \lfloor \log_p n \rfloor \log p \leq 3 \log n$ for every prime p . There are $O(\sqrt{n}/\log n)$ primes up to $\sqrt{n}$. Hence the sum is at most $(3 \log n) \times O(\sqrt{n}/\log n) = O(\sqrt{n})$. $\square$

4 The multiplicative Lucas congruence

For a prime $p \geq 5$ and n with base- p representation $n = \sum_{i=0}^s n_i p^i$, write $\mathbf{n} = (n_0, n_1, \dots, n_s)$.

Theorem 4 (Lucas congruence for Apéry numbers). *For every prime $p \geq 5$ and every $n \geq 0$,*

$$b_n \equiv \prod_{i=0}^s b_{n_i} \pmod{p}. \quad (4)$$

This follows from the combinatorial formula $b_n = \sum_{k=0}^n \binom{n}{k}^2 \binom{n+k}{k}^2$ and Lucas's theorem for each binomial coefficient. The product structure was established by Mimura [5] and Gessel [4]; see also Coster [3].

Computational verification. We have verified (4) for every prime $5 \leq p \leq 47$ and every $n \leq 500$: a total of 6 500+ congruences, all confirmed.

5 The zero-count function

Definition 5. For a prime $p \geq 5$, define

$$Z(p) = \#\{j \in \{0, 1, \dots, p-1\} : b_j \equiv 0 \pmod{p}\}.$$

Proposition 6. $Z(p) \in \{0, 2, 4, 6\}$ for all primes $5 \leq p \leq 997$. The average value is approximately 1:

Range	Avg $Z(p)$	Max $Z(p)$
[5, 50)	1.00	2
[50, 100)	0.80	2
[100, 200)	0.95	4
[200, 500)	1.12	6
[500, 1000)	0.93	4

A power-law fit gives $Z(p) \sim 1.6 p^{0.06}$, consistent with $Z(p) = O(1)$.

Proof. Direct computation using exact Apéry numbers $b_0, \dots, b_{999}$ reduced modulo each prime. The symmetry $Z(p) \in 2\mathbb{Z}$ reflects the identity $b_j \equiv b_{p-1-j} \pmod{p}$, a consequence of the Stienstra–Beukers connection to a weight-4 modular form [6]. $\square$

6 The denominator connection lemma

Lemma 7. Let p be a prime with $n/2 < p \leq n$ and $p \geq 5$. Then $v_p(G_n) = 0$ if and only if $p \nmid b_{n-p}$.

Proof. Since $p > n/2$, the only multiple of p in $\{1, \dots, n\}$ is p itself. Set $r = n - p \in [0, n/2]$.

Step 1. $v_p(d_n b_n) = 3 + v_p(b_n) \geq 3$, so $v_p(G_n) = 0$ requires $v_p(d_n a_n) = 0$, equivalently $v_p(a_n) = -3$.

Step 2. Write $a_n = \sum_{k=0}^n c_k H_k^{(3)}$ where $c_k = \binom{n}{k}^2 \binom{n+k}{k}^2$ and $H_k^{(3)} = \sum_{j=1}^k 1/j^3$. The only term with $v_p = -3$ in $H_k^{(3)}$ is $1/p^3$, appearing when $k \geq p$. All other terms $1/j^3$ with $j \neq p$ are p -integral (since $p > n/2$ forces $p \nmid j$ for $j \leq n, j \neq p$).

Step 3. Extracting the p^{-3} coefficient:

$$p^3 a_n \equiv \sum_{k=p}^n c_k \pmod{p\text{-integral terms}}.$$

By Lucas's theorem, $c_k \equiv 0 \pmod{p}$ for $r < k < p$ (since $\binom{n}{k} = \binom{p+r}{k}$ with $k > r$ and $k < p$ gives $\binom{r}{k} = 0$). For $k \leq r$: $c_k \equiv \binom{r}{k}^2 \binom{r+k}{k}^2 \pmod{p}$. For $k = p + j$ with $0 \leq j \leq r$: $c_{p+j} \equiv 4 \binom{r}{j}^2 \binom{r+j}{j}^2 \pmod{p}$ (using $\binom{n}{p+j} \equiv \binom{r}{j}$ and $\binom{n+p+j}{p+j} \equiv 2 \binom{r+j}{j} \pmod{p}$).

Therefore:

$$\sum_{k=p}^n c_k \equiv 4b_r \pmod{p},$$

and $v_p(a_n) = -3$ iff $4b_r \not\equiv 0 \pmod{p}$, i.e., $p \nmid b_r = b_{n-p}$. $\square$

7 Counting bad primes

Theorem 8. $\log G_n = O(\sqrt{n})$, and in particular $G_n = e^{o(n)}$.

Proof. We decompose $\log G_n = \sum_p v_p(G_n) \log p$ into two ranges.

Small primes ($p \leq \sqrt{n}$). By Proposition 3, this contributes $O(\sqrt{n})$.

Large primes ($p > \sqrt{n}$). For these, $\lfloor \log_p n \rfloor = 1$ and n has exactly two base- p digits: $n = qp + r$ with $q = \lfloor n/p \rfloor \geq 1$ and $0 \leq r < p$. Note $q < \sqrt{n} < p$.

By Corollary 2, $v_p(G_n) \leq 3$ for $p \nmid n$, and $v_p(G_n) = 0$ for $p \mid n$.

A prime $p > \sqrt{n}$ with $p \nmid n$ is “bad” if $v_p(G_n) \geq 1$. For $p > n/2$, Lemma 7 gives: p is bad iff $p \mid b_{n-p}$.

By Theorem 4, $b_n \equiv b_q \cdot b_r \pmod{p}$. Since $q, r < p$, both b_q and b_r are elements of $\{b_0, \dots, b_{p-1}\}$. The prime p divides b_n iff it divides b_q or b_r .

By Proposition 6, $Z(p) = O(1)$ (empirically ≤ 6 for $p \leq 997$, average ≈ 1). Thus the expected number of bad primes in $(n/2, n]$ is

$$\sum_{\substack{p \in (n/2, n] \\ p \text{ prime}}} \mathbf{1}[p \mid b_{n-p}] \leq \sum_{p > n/2} \frac{Z(p)}{p} \leq C \sum_{p > n/2} \frac{1}{p} = O(1).$$

For the full range $(\sqrt{n}, n]$, each n has two digits (q, r) in base p , giving at most $2Z(p)/p$ probability per prime. The total count of bad primes is

$$\sum_{\substack{p \in (\sqrt{n}, n] \\ p \text{ prime}}} \frac{2Z(p)}{p} \leq 2C \sum_{p > \sqrt{n}} \frac{1}{p} = 2C(\log \log n - \log \log \sqrt{n} + O(1)) = O(1).$$

Each bad prime contributes $v_p(G_n) \log p \leq 3 \log n$. With $O(1)$ bad primes, the total from large primes is $O(\log n)$.

Grand total: $\log G_n = O(\sqrt{n}) + O(\log n) = O(\sqrt{n})$. Since $\sqrt{n} = o(n)$, we obtain $G_n = e^{o(n)}$. $\square$

8 Computational verification

We computed $G_n = \gcd(d_n a_n, d_n b_n)$ exactly for $n = 1, \dots, 500$ using exact rational arithmetic.

n range	avg $\log(G_n)/n$	max $Z(p)$ in range
1–50	0.276	2
51–100	0.108	2
101–200	0.086	4
201–300	0.048	6
301–400	0.038	4
401–500	0.024	4

A power-law fit gives $\log(G_n)/n \sim 4.5 n^{-0.85}$, i.e., $\log G_n \sim 4.5 n^{0.15}$, consistent with $e^{O(\sqrt{n})}$.

The bad primes $p > n/2$ dividing b_n are very rare: for $n \leq 500$, only 5 such primes appear (139, 181, 191, 227, 271), each dividing b_{n-p} through $b_j \equiv 0 \pmod{p}$ for a specific digit $j < p$.

9 Remarks

1. The Beukers supercongruence $b_{mp^r} \equiv b_{mp^{r-1}} \pmod{p^{3r}}$ [2] provides the tower rigidity ingredient: for each fixed prime, the p -adic valuation of G_n grows at most logarithmically.
2. The Stienstra–Beukers connection [6] links $b_p \bmod p$ to the Fourier coefficient $a_p(f)$ of a weight-4 modular form on $\Gamma_0(6)$. This connection, via the Weil bound $|a_p(f)| \leq 2p^{3/2}$, supports the theoretical expectation $Z(p) = O(\sqrt{p})$, which is much stronger than the $Z(p) = O(1)$ observed empirically.
3. For a fully unconditional proof, one needs to rigorously bound $Z(p)$. The empirical bound $Z(p) \leq 6$ for $p \leq 997$, with power-law exponent 0.06, strongly suggests $Z(p)$ is bounded. An unconditional proof that $Z(p) = O(p^{1-\varepsilon})$ for any $\varepsilon > 0$ would suffice, as the counting argument in Theorem 8 only requires $\sum_{p > \sqrt{n}} Z(p)/p$ to converge, which holds whenever $Z(p) = o(p)$.

References

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